Data Dependence Analysis by Entropy-Based Quantities

Daniel Chou[1]

[1]Georgia Institute of Technology, School of Civil and Environmental Engineering, Atlanta, GA 30332, United States of America

ABSTRACT

The concept of entropy from information theory is utilized to quantify the uncertainty of a system. Recognizing that most introductory texts on this topic present repeated and sometimes ambiguous mathematical definitions, this report aims to offer clear and comprehensive illustrations. Initially, it outlines a standard process for calculating entropy-related quantities for continuous variables. This algorithm employs Kernel Density Estimation (KDE) and Gauss-Legendre Quadrature, building upon the original mathematical definitions.
Subsequently, experiments regarding entropy-based measurements, such as mutual information, are performed to validate the accuracy of entropy estimation results. Additionally, the concept of Transfer Entropy (TE) is introduced, accompanied by extensive experiments to confirm TE's efficacy in quantifying information transfer between variables.
The conclusions highlight the essential components and limitations of TE. Codes could be downloaded at github.

1 Introduction to Entropy

The entropy is used to quantify the system uncertainty. Several entropy-related definitions of continuous variable are given in this section.

1. **Marginal Entropy (Continuous):**
   - The uncertainty of a continuous random variable.
   - $H(X) = -\int_{-\infty}^{\infty} p(x)\log p(x)\,dx$
2. **Joint Entropy (Continuous):**
   - The uncertainty of a joint system of two continuous variables.
   - $H(X,Y) = -\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x,y)\log p(x,y)\,dy\,dx$
3. **Conditional Entropy (Continuous):**
   - The uncertainty of a continuous variable given another.
   - $H(Y|X) = -\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x,y)\log \frac{p(x,y)}{p(x)}\,dy\,dx$
   - $H(Y|X) = H(X,Y) - H(X)$.
4. **Mutual Information (Continuous):**
   - The amount of information that one continuous variable contains about another. It measures the reduction in uncertainty about one variable given the knowledge of another.
   - $I(X;Y) = H(X) + H(Y) - H(X,Y)$
   - Alternatively, $I(X;Y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x,y)\log \frac{p(x,y)}{p(x)p(y)}\,dx\,dy$
   - Mutual Information is always non-negative and quantifies the shared information between variables. It is zero if and only if the variables are independent.

Suppose we have two random variables, x, y, estimating these quantities at least needs the marginal entropy and the joint entropy. The direct calculation process usually contains Kernel Density Estimation(KDE) to obtain the probability density, e.g., $p(x)$, $p(y)$ and $p(x,y)$, furthermore, the numerical integration is required, i.e., to deal with $\int p(x) \log p(x) dx$. The details are given in section 2.

2 Algorithm Outline

This section provides an outline to entropy quantities estimation, the example of two features case is used, and the goal is to calculate the marginal entropy, joint entropy and the mutual information. Suppose we have two features in one array with N data points, denoted as $\mathbf{J} = [\mathbf{x}, \mathbf{y}] \in \mathbb{R}^{N\times 2}$, where both $\mathbf{x}, \mathbf{y}$ are column vector with shape $N \times 1$. The works flow of the entire process is given in Figure 1.

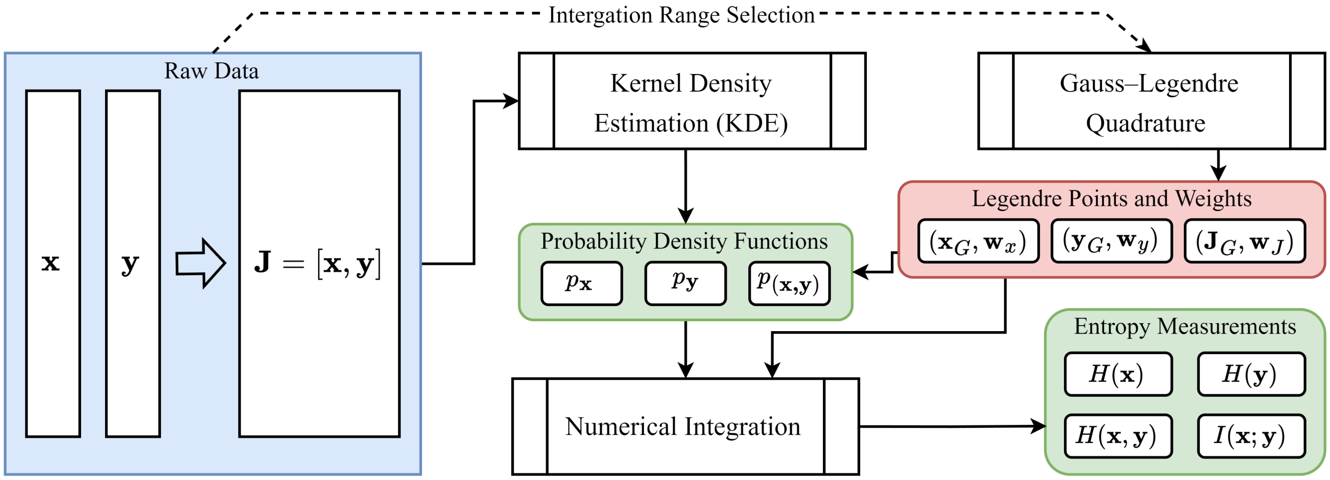


Figure 1. The Pipe Line of Entropy Estimation

And the consecutive steps are listed as follows:

1. First, use Kernel Density Estimation (KDE) to obtain the probability density functions for two features, denoted as $p_\mathbf{x}$ for $\mathbf{x}$, $p_\mathbf{y}$ for $\mathbf{y}$, and $p_{(\mathbf{x},\mathbf{y})}$ for the joint distribution of $\mathbf{x}$ and $\mathbf{y}$. Here, $\mathbf{x}$ and $\mathbf{y}$ are column vectors within the data array $\mathbf{J} = [\mathbf{x}, \mathbf{y}] \in \mathbb{R}^{N\times 2}$, where N is the number of data points.

2. Next, generate Gauss Legendre quadrature points and weights for $\mathbf{x}$, denoted as $(\mathbf{x}_G, \mathbf{w}_x)$, for $\mathbf{y}$, denoted as $(\mathbf{y}_G, \mathbf{w}_y)$, and for the joint distribution $(\mathbf{J}_G, \mathbf{w}_J)$. The total number of points is defined by n_x, n_y, and $n_J = n_x \times n_y$, respectively. The dependence between the joint Gauss Legendre points and weights and the individual points and weights is given by:

$$\begin{aligned}
&\mathbf{J}_{G(ij)} = [\mathbf{x}_{G(i)}, \mathbf{y}_{G(j)}],\\
&\mathbf{w}_{J(ij)} = \mathbf{w}_{x(i)} \cdot \mathbf{w}_{y(j)}.
\end{aligned}$$

   These are reshaped to:

$$\begin{aligned}
&\mathbf{J}_G \in \mathbb{R}^{n_x\times n_y\times 2} \to \mathbf{J}_G \in \mathbb{R}^{n_J\times 2},\\
&\mathbf{w}_J \in \mathbb{R}^{n_x\times n_y} \to \mathbf{w}_J \in \mathbb{R}^{n_J}.
\end{aligned}$$

3. Finally, calculate the entropy for $\mathbf{x}$, $\mathbf{y}$, and their joint distribution using Gauss Legendre numerical integration:

$$\begin{aligned}
H(\mathbf{x}) &= -\sum_{i=1}^{n_x} \mathbf{w}_{x(i)} \cdot p_\mathbf{x}(\mathbf{x}_{G(i)}) \cdot \log p_\mathbf{x}(\mathbf{x}_{G(i)}),\\
H(\mathbf{y}) &= -\sum_{i=1}^{n_y} \mathbf{w}_{y(i)} \cdot p_\mathbf{y}(\mathbf{y}_{G(i)}) \cdot \log p_\mathbf{y}(\mathbf{y}_{G(i)}),\\
H(\mathbf{x},\mathbf{y}) &= -\sum_{i=1}^{n_J} \mathbf{w}_{J(i)} \cdot p_{(\mathbf{x},\mathbf{y})}(\mathbf{J}_{G(i)}) \cdot \log p_{(\mathbf{x},\mathbf{y})}(\mathbf{J}_{G(i)}),\\
I(\mathbf{x};\mathbf{y}) &= \sum_{i=1}^{n_J} \mathbf{w}_{J(i)} \cdot p_{(\mathbf{x},\mathbf{y})}(\mathbf{J}_{G(i)}) \cdot \left(\log p_{(\mathbf{x},\mathbf{y})}(\mathbf{J}_{G(i)}) - \log p_\mathbf{x}(\mathbf{x}_{G(i)}) - p_\mathbf{y}(\mathbf{y}_{G(i)})\right)\\
&= H(\mathbf{x}) + H(\mathbf{y}) - H(\mathbf{x},\mathbf{y}).
\end{aligned}$$

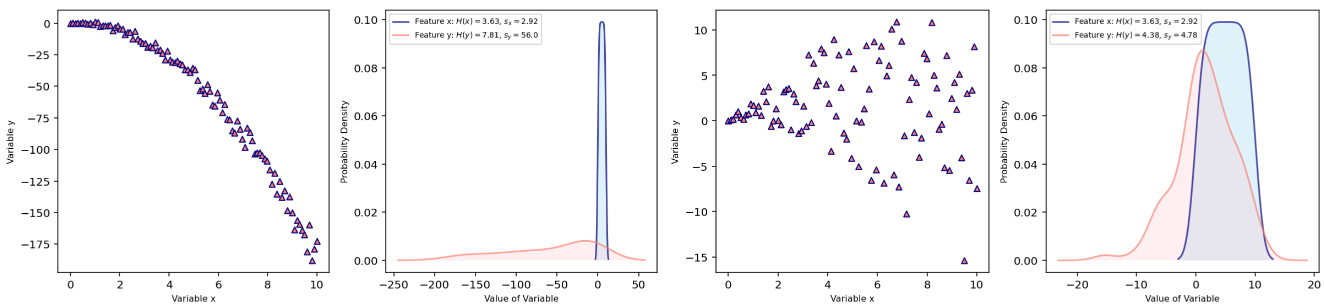
3 Exploration of Entropy Related Properties

3.1 Marginal Entropy

The marginal entropy is used to quantify the uncertainty of a random variable. In this section, a stationary feature vector $\mathbf{x} \in \mathbb{R}^{100}$ is generated by a linear-spacing function in the range $[0, 10]$, i.e., $\mathbf{x} = \{0, 0.1, 0.2, \dots, 10\}$. Another feature $\mathbf{y}$ is generated based on $\mathbf{x}$ using a different approach. The marginal entropy of the two features is observed, along with their standard deviations, to support the explanation of the results.

1. Case 1: $y_i = -2x_i^2 + \varepsilon \cdot z_i$
2. Case 2: $y_i = -0.2x_i^2 + \varepsilon \cdot z_i$

Where ε is a random noise term uniformly distributed, such that $\varepsilon \in [0, 1]$. z_i is a linearly increasing scaling factor for the noise, ranging from 0 to 30 over the number of data points, effectively making the noise heteroscedastic, i.e., the variability of the noise increases with x. The results are presented in Figure 2. In Case 1 (Figure 2a), the standard deviation of $\mathbf{y}$ is significantly larger than in Case 2 (Figure 2b). Furthermore, the entropy $H(y)$ from Case 2 is also higher than that from Case 1. This indicates that spreading the samples over a wider range increases the uncertainty of the random variable, as a larger standard deviation implies greater difficulty in predicting the location of the next new sample, thereby increasing the uncertainty.



(a) Case 1 **(b)** Case 2

Figure 2. Marginal Entropy of Both Cases

3.2 Feature Dependence Evaluation by Mutual Information

Mutual information is a non-parametric measure of the dependency between two variables[1]. It quantifies the amount of information one variable contains about the other, regardless of the correlation type (linear or non-linear) and without assuming any specific data distribution. Since the process involves numerical estimation, two approaches are used to calculate the mutual information:

1. Direct calculation using marginal entropy and joint entropy.

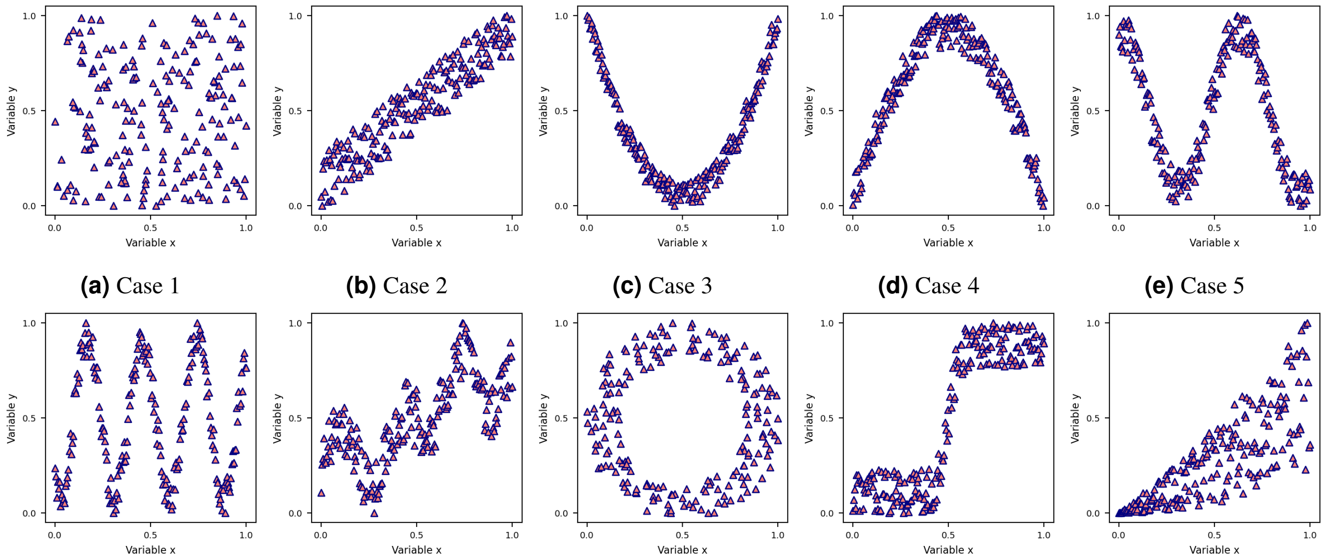
$$I_1(\mathbf{x};\mathbf{y}) = H(\mathbf{x}) + H(\mathbf{y}) - H(\mathbf{x},\mathbf{y}).$$

2. Direct numerical integration based on the probability density functions.

$$I_2(\mathbf{x};\mathbf{y}) = \sum_{i=1}^{n_J} \mathbf{w}_{J(i)} \cdot p_{(\mathbf{x},\mathbf{y})}(\mathbf{J}_{G(i)}) \cdot \left(\log p_{(\mathbf{x},\mathbf{y})}(\mathbf{J}_{G(i)}) - \log p_{\mathbf{x}}(\mathbf{x}_{G(i)}) - \log p_{\mathbf{y}}(\mathbf{y}_{G(i)})\right).$$

Ten cases are presented in Figure 3, and their mutual information is listed in Table 1. Pearson's correlation coefficient $\rho_{(\mathbf{x},\mathbf{y})}$ and the distance correlation coefficient $\mathrm{dCor}(\mathbf{x},\mathbf{y})$ are also provided.

It can be observed that calculating mutual information directly from marginal and joint entropies, as in $I_1(\mathbf{x};\mathbf{y}) = H(\mathbf{x}) + H(\mathbf{y}) - H(\mathbf{x},\mathbf{y})$, is more likely to yield negative values. However, mutual information should inherently be non-negative. This issue likely stems from the amplification of numerical errors when estimating the marginal entropies $H(\mathbf{x})$, $H(\mathbf{y})$, and the joint entropy $H(\mathbf{x},\mathbf{y})$ separately. This problem is not merely due to the lack of a guaranteed hierarchical relationship among these entropy quantities in the algorithms used, but also because, for continuous variables, the entropy (so-called differential entropy in continuous) hierarchy that exists for discrete variables does not apply. **Therefore, a direct numerical integration approach is adopted in the remainder of this report.**



(a) Case 1 **(b)** Case 2 **(c)** Case 3 **(d)** Case 4 **(e)** Case 5

(f) Case 6 **(g)** Case 7 **(h)** Case 8 **(i)** Case 9 **(j)** Case 10

Figure 3. Mutual Information Tests

Data	$H(\mathbf{x})$	$H(\mathbf{y})$	$H(\mathbf{x},\mathbf{y})$	$I_1(\mathbf{x};\mathbf{y})$	$I_2(\mathbf{x};\mathbf{y})$	$\rho_{(\mathbf{x},\mathbf{y})}$	dCor$(\mathbf{x},\mathbf{y})$
Case 1	0.217	0.273	0.578	-0.089	0.021	0.05	0.086
Case 2	0.268	0.077	-1.135	1.479	1.573	0.94	0.932
Case 3	0.268	0.149	0.077	0.339	0.46	-0.03	0.486
Case 4	0.268	0.197	0.115	0.35	0.46	0.01	0.486
Case 5	0.268	0.294	0.409	0.153	0.277	-0.24	0.354
Case 6	0.268	0.267	0.624	-0.089	0.03	0.02	0.159
Case 7	0.268	-0.074	-0.283	0.477	0.568	0.69	0.739
Case 8	0.275	0.282	0.536	0.021	0.156	-0.01	0.168
Case 9	0.268	0.341	-0.3	0.908	1.07	0.88	0.91
Case 10	0.268	-0.064	-0.622	0.825	0.937	0.81	0.806

Table 1. Entropy, Mutual Information and Correlation Coefficients

3.2.1 Summary for Mutual Information

In summary, mutual information(MI) is an advantageous metric for measuring the dependence between variables for several reasons:

1. MI is a non-parametric metric, meaning it does not rely on deriving any statistical parameters (such as mean or standard deviation). Therefore, it allows for the evaluation of correlations without assuming any specific data distribution. For instance, using the mean and standard deviation typically implies an assumption of Gaussian distribution, which is not required here. Furthermore, MI can capture the underlying correlation between features, regardless of the type of correlation (linear or non-linear). This makes it sufficiently general for most cases.

2. MI can be calculated without restrictions on data structure, consider two arrays $\mathbf{a} = [\mathbf{a_1}, \mathbf{a_2}]$ and $\mathbf{b} = [\mathbf{b_1}]$, where $\mathbf{a_i}$ and $\mathbf{b_j}$ are $N \times 1$ column features. The traditional correlation coefficients require the two variables to have the same dimensionality, e.g., for Pearson's correlation, only $\rho(\mathbf{a_i}, \mathbf{b_j})$ are available. Traditional correlation coefficient does not offer the approach to explore the dependence between the joint array $\mathbf{a} = [\mathbf{a_1}, \mathbf{a_2}]$ and $\mathbf{b}$. By treating $\mathbf{a}$ as a single entity and comparing it with $\mathbf{b}$, MI allows for the analysis of how the combination of $\mathbf{a_1}$ and $\mathbf{a_2}$ together predicts or informs about $\mathbf{b_1}$, capturing more complex, possibly nonlinear relationships that might not be apparent when looking at $\mathbf{a_1}$ and $\mathbf{b_1}$ or $\mathbf{a_2}$ and $\mathbf{b_1}$ in isolation.

However, the drawbacks of using mutual information for capturing correlation are also quite evident:

1. It presents a steep learning curve, especially regarding implementation, as there is scant practical guidance available. The literature often lacks straightforward examples, making it challenging for beginners to grasp. Most sources tend to repeat the mathematical formulas without offering concrete examples, significantly increasing the learning barrier.

2. For continuous variables, the calculation involves Kernel Density Estimation (KDE) and numerical integration, adding complexity to the process. Moreover, the logarithmic operations can cause instability when density values are near zero.

3. In the case of discrete variables, the outcome can be highly sensitive to the chosen number of bins. An inappropriate selection can lead to inaccurate or even meaningless results.

4. Mutual information cannot be directly incorporated into Deep Learning objective functions because the general estimation process, as shown in Figure 1, is not differentiable. The numerical integration involves Gauss points generated independently of the original inputs and model parameters, making it impossible to directly obtain $\frac{\partial(I(\mathbf{x};\mathbf{y}))}{\partial\theta}$, where θ represents the parameters being updated in Deep Learning model. Alternatives like the Mutual Information Neural Estimator (MINE)[2,3] are required.

5. As indicated in Table 1, simpler metrics such as Pearson's correlation and distance correlation might suffice for quantifying feature dependence. In some cases, mutual information performs worse than these simpler metrics. This is observed in case 6, as shown in Figure 3f and Table 1. Although a sine correlation between features exists, mutual information $I_2(\mathbf{x};\mathbf{y}) = 0.03$ indicates a very small correlation, whereas the distance correlation $\text{dCor}(\mathbf{x},\mathbf{y}) = 0.159$ successfully identifies the correlation.

To conclude, mutual information is a valuable metric with high flexibility and a solid theoretical foundation for evaluating variable dependence. However, it also has limitations that can make it less popular than more intuitive measures like Pearson's or distance correlations.

3.2.2 Scale Independence

Similar to correlation coefficients, mutual information is scale-independent; that is, it does not change with the scaling of data. Figure 4 displays data points in the original, max-min normalized, and z-score normalized scales, showcasing a quadratic correlation between variables. The corresponding values of mutual information are presented in Figure 2. As demonstrated in Figure 2, mutual information is indeed a scale-invariant quantity.

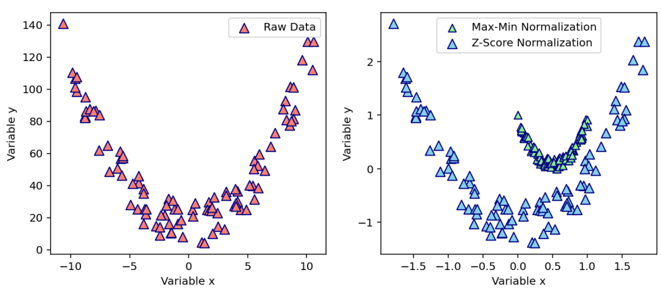


Figure 4. Comparison of normalization techniques

Data Scale	$I(\mathbf{x};\mathbf{y})$
Original Scale	0.37282
Max-Min Normalization	0.37282
Z-Score Normalization	0.37282

Table 2. Mutual Information Values

3.3 Entropy Approximation by Covariance Matrix

As indicated in recent studies[4–6], entropy for Gaussian random variable can be approximated using the covariance matrix of data. Under this approximation, the entropy and joint entropy are expressed as:

$$H(\cdot) = \frac{k}{2}\log(2\pi e) + \frac{1}{2}\log(\det(\Sigma_{(\cdot)})),$$

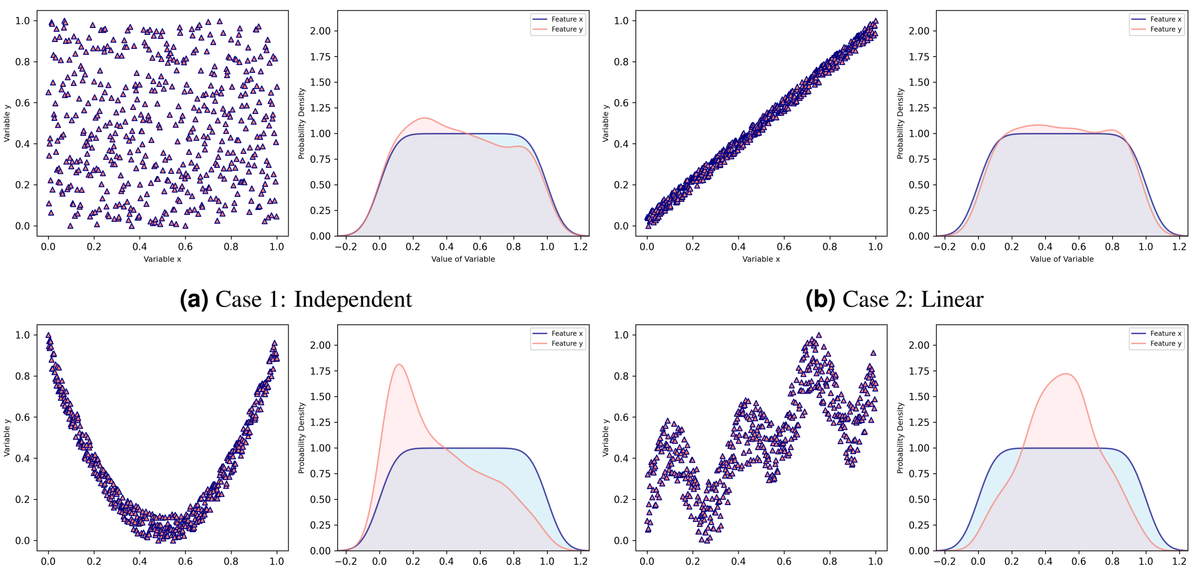
where $(\cdot)$ represents arbitrary random variable data, which may take the form of an array or a matrix. Here, k is the number of features of the data, and $\Sigma_{(\cdot)}$ is its covariance matrix.

An experiment was conducted with variables $\mathbf{x}$ and $\mathbf{y}$. Their marginal entropy, joint entropy, and mutual information were calculated using both numerical integration and covariance matrix approximation methods. To quantify the differences between these two approaches, the percentage difference was calculated as follows:

$$p_d = \frac{|Q_n - Q_a|}{Q_n} \times 100\%,$$

where Q_n represents the quantity estimated by numerical integration, which is assumed to be closer to the true value as it adheres more closely to the original mathematical definitions without significant simplification. Q_a denotes the quantity derived from the covariance approximation.

The experimental scenarios, including cases with independent, linear, non-linear, and semi-linear correlation between $\mathbf{x}$ and $\mathbf{y}$, are illustrated in Figure 5. The results from these experiments are documented in Table 3.



(a) Case 1: Independent **(b)** Case 2: Linear

(c) Case 3: Non-linear **(d)** Case 4: Semi-Linear

Figure 5. Test cases illustrating various correlation between $\mathbf{x}$ and $\mathbf{y}$ and their impact on entropy approximations.

The insights gleaned from Table 3 lead to the following conclusions:

1. Concerning marginal entropy, when the data distribution is inherently Gaussian (bell-shaped), the errors are in the range of 10% to 30%, as illustrated in Figure 5a, Figure 5b, and Figure 5d. However, for variable $\mathbf{y}$ in Figure 5c, which has a highly skewed distribution, the error soars to 137.73%, indicating a significant discrepancy. In scenarios with linear or semi-linear correlation (Figure 5b and Figure 5d), the covariance approximation method estimates mutual information quite effectively.

2. Regarding joint entropy and mutual information, in the independent case (Figure 5a), although the error of mutual information reaches 61.93%, the $I(\mathbf{x};\mathbf{y})$ values from both methods are quite low. The covariance approximation ($I(\mathbf{x};\mathbf{y}) = 0.006$) performs better than numerical integration ($I(\mathbf{x};\mathbf{y}) = 0.016$) in indicating independence. This shows that in certain scenarios, specifically when the raw data are Gaussian variables and are independent, the covariance approximation evaluates mutual information more accurately than numerical integration. However, when variables

Case	Method	$H(\mathbf{x})$	$H(\mathbf{y})$	$H(\mathbf{x},\mathbf{y})$	$I(\mathbf{x};\mathbf{y})$
Case 1	Numerical Integration	0.201	0.19	0.465	0.016
	Covariance Approximation	0.235	0.213	0.442	0.006
	Percentage Difference	16.7%	12.15%	4.95%	61.93%
Case 2	Numerical Integration	0.22	0.14	-2.909	3.324
	Covariance Approximation	0.259	0.176	-2.927	3.362
	Percentage Difference	17.66%	25.95%	0.62%	1.14%
Case 3	Numerical Integration	0.22	0.092	-0.151	0.53
	Covariance Approximation	0.259	0.219	0.478	0.0
	Percentage Difference	17.66%	137.73%	416.09%	100.0%
Case 4	Numerical Integration	0.22	-0.155	-0.444	0.553
	Covariance Approximation	0.259	-0.201	-0.413	0.471
	Percentage Difference	17.66%	29.16%	6.94%	14.71%

Table 3. Entropy Approximation Test for Multiple Cases

exhibit a non-linear correlation, the errors in entropy estimation by covariance become unacceptably high. For example, in Figure 5c, mutual information is indicated as zero from covariance approximation, even when there is a clear quadratic correlation between the variables.

These findings point out that while the covariance approximation method for entropy is useful, it has its limitations. Acceptable results are typically achieved under the assumptions that the data conforms to a Gaussian distribution and that there is a linear (or at least semi-linear) correlation between variables.

4 Transfer Entropy

Transfer Entropy (TE) is a concept from information theory that quantifies the influence one data sequence has over another[7,8].

4.1 Theoretical Framework and Mathematical Definitions

Given a source sequential array $\mathbf{x}$ and a target sequential array $\mathbf{y}$, each with length T:

$$
\begin{aligned}
\mathbf{x} &= \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T\},\\
\mathbf{y} &= \{\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_T\},
\end{aligned}
$$

where $\mathbf{x}$ and $\mathbf{y}$ are in the shape $\mathbb{R}^{N\times T}$, representing N observations at each time step. $\mathbf{x}_t$ and $\mathbf{y}_t$ are 1-D arrays within the shape $\mathbb{R}^{N\times 1}$. For a specific time step t, the sub-sequences including past states up to the current time are defined by the history length τ:

$$
\begin{aligned}
\mathbf{x}_t^\tau &= \{\mathbf{x}_{t-\tau+1}, \ldots, \mathbf{x}_t\},\\
\mathbf{y}_t^\tau &= \{\mathbf{y}_{t-\tau+1}, \ldots, \mathbf{y}_t\}.
\end{aligned}
$$

Sub-sequences including future steps are detailed by h:

$$
\begin{aligned}
\mathbf{x}_{t+h} &= \{\mathbf{x}_{t+1}, \ldots, \mathbf{x}_{t+h}\},\\
\mathbf{y}_{t+h} &= \{\mathbf{y}_{t+1}, \ldots, \mathbf{y}_{t+h}\}.
\end{aligned}
$$

The uncertainty of future values of $\mathbf{y}_{t+h}$ given the past and current values $\mathbf{y}_t^\tau$ is quantified by the conditional entropy $H(\mathbf{y}_{t+h}|\mathbf{y}_t^\tau)$. The transfer entropy from $\mathbf{x}$ to $\mathbf{y}$ at time t for future window h is the reduction in uncertainty about $\mathbf{y}_{t+h}$ when past values from $\mathbf{x}_t^\tau$ are also considered:

$$
\begin{aligned}
T_{\mathbf{x}\to\mathbf{y}}(t,\tau,h) &= H(\mathbf{y}_{t+h}|\mathbf{y}_t^\tau) - H(\mathbf{y}_{t+h}|\mathbf{y}_t^\tau,\mathbf{x}_t^\tau)\\
&= H(\mathbf{y}_{t+h},\mathbf{y}_t^\tau) - H(\mathbf{y}_t^\tau) - H(\mathbf{y}_{t+h},\mathbf{y}_t^\tau,\mathbf{x}_t^\tau) + H(\mathbf{y}_t^\tau,\mathbf{x}_t^\tau)
\end{aligned} \tag{1}
$$

Figure 6a provides a conceptual overview for transfer entropy. Alternatively, the transfer entropy can also be viewed as the conditional mutual information between $\mathbf{y}_{t+h}$ and $\mathbf{x}_t^\tau$, given $\mathbf{y}_t^\tau$, as outlined by Shahsavari et al.[8]:

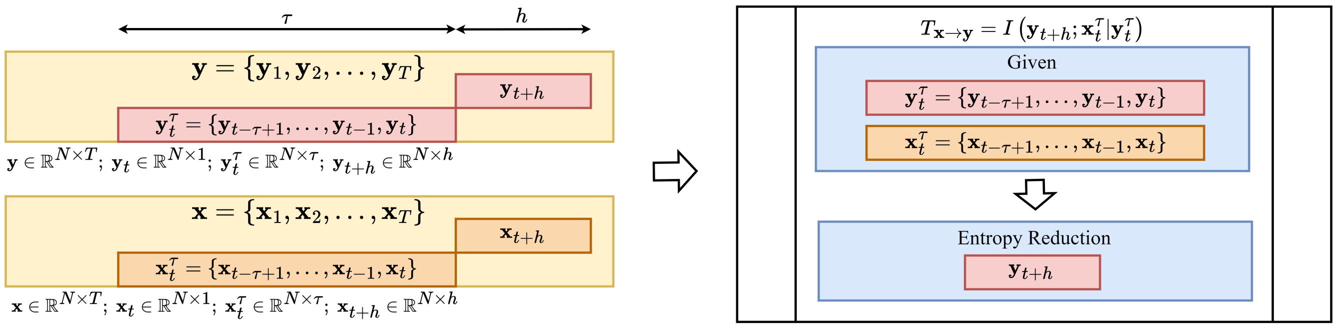
$$
\begin{aligned}
T_{\mathbf{x}\to\mathbf{y}}(t,\tau,h) &= I(\mathbf{y}_{t+h};\mathbf{x}_t^\tau|\mathbf{y}_t^\tau)\\
&= \int_{\mathbf{y}_{t+h}}\int_{\mathbf{x}_t^\tau}\int_{\mathbf{y}_t^\tau} p(\mathbf{y}_{t+h},\mathbf{x}_t^\tau,\mathbf{y}_t^\tau)\log\left(\frac{p(\mathbf{y}_t^\tau)p(\mathbf{y}_{t+h},\mathbf{x}_t^\tau,\mathbf{y}_t^\tau)}{p(\mathbf{y}_{t+h},\mathbf{y}_t^\tau)p(\mathbf{x}_t^\tau,\mathbf{y}_t^\tau)}\right) d\mathbf{y}_{t+h}d\mathbf{x}_t^\tau d\mathbf{y}_t^\tau
\end{aligned} \tag{2}
$$

For simplicity, the future window h is set to one step ahead, i.e., $h=1$. In this case, the expressions in Equation 1 and Equation 2 simplify as follows:

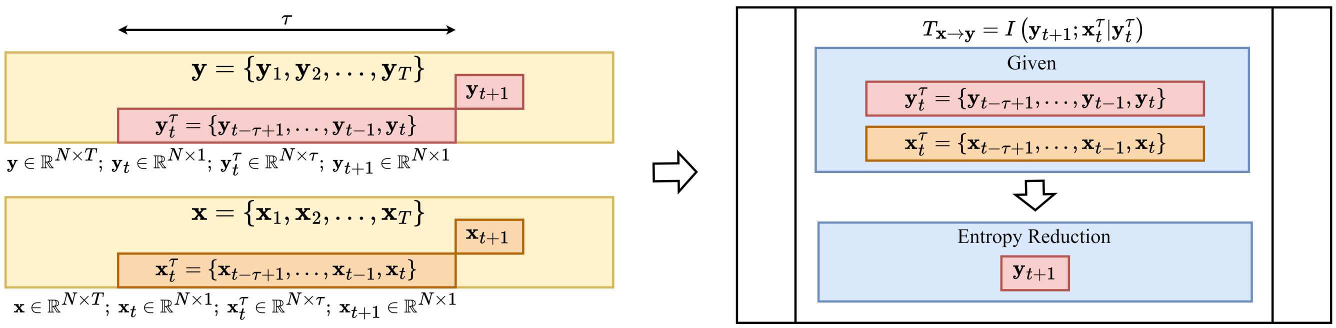
$$
T_{\mathbf{x}\to\mathbf{y}}(t,\tau) = H(\mathbf{y}_{t+1},\mathbf{y}_t^\tau) - H(\mathbf{y}_t^\tau) - H(\mathbf{y}_{t+1},\mathbf{y}_t^\tau,\mathbf{x}_t^\tau) + H(\mathbf{y}_t^\tau,\mathbf{x}_t^\tau) \tag{3}
$$

$$
\begin{aligned}
&= \int_{\mathbf{y}_{t+1}}\int_{\mathbf{x}_t^\tau}\int_{\mathbf{y}_t^\tau} p(\mathbf{y}_{t+1},\mathbf{x}_t^\tau,\mathbf{y}_t^\tau)\log\left(\frac{p(\mathbf{y}_t^\tau)p(\mathbf{y}_{t+1},\mathbf{x}_t^\tau,\mathbf{y}_t^\tau)}{p(\mathbf{y}_{t+1},\mathbf{y}_t^\tau)p(\mathbf{x}_t^\tau,\mathbf{y}_t^\tau)}\right) d\mathbf{y}_{t+1}d\mathbf{x}_t^\tau d\mathbf{y}_t^\tau\\
&= \int_{\mathbf{y}_{t+1}}\int_{\mathbf{x}_t^\tau}\int_{\mathbf{y}_t^\tau} p(\mathbf{y}_{t+1},\mathbf{x}_t^\tau,\mathbf{y}_t^\tau)\left\{\log(p(\mathbf{y}_t^\tau)) + \log(p(\mathbf{y}_{t+1},\mathbf{x}_t^\tau,\mathbf{y}_t^\tau)) - \log(p(\mathbf{y}_{t+1},\mathbf{y}_t^\tau)) - \log(p(\mathbf{x}_t^\tau,\mathbf{y}_t^\tau))\right\} d\mathbf{y}_{t+1}d\mathbf{x}_t^\tau d\mathbf{y}_t^\tau
\end{aligned} \tag{4}
$$

Equation 3 indicates that calculating transfer entropy involves only marginal and joint entropies, which can be readily estimated using the process outlined in Figure 1. Figure 6b provides the conceptual overview of transfer entropy when $h=1$, and Equation 4 provides a more stable approach to calculating transfer entropy through direct numerical integration. In contrast, as shown in Equation 3, when joint entropy terms are calculated individually, numerical errors are likely to accumulate during the process. Furthermore, there is no guarantee that the transfer entropy will always be greater than or equal to zero, though the mutual information should not be negative. The transfer entropy estimation process by using direct numerical integration in Equation 4 is given in Figure 7. And as described previously in subsubsection 3.2.2, transfer entropy is also data scale independent, as the mutual information value should be independent to the scale of data.



(a) General Case



(b) Case of One-Step Ahead

Figure 6. Concept of Transfer Entropy

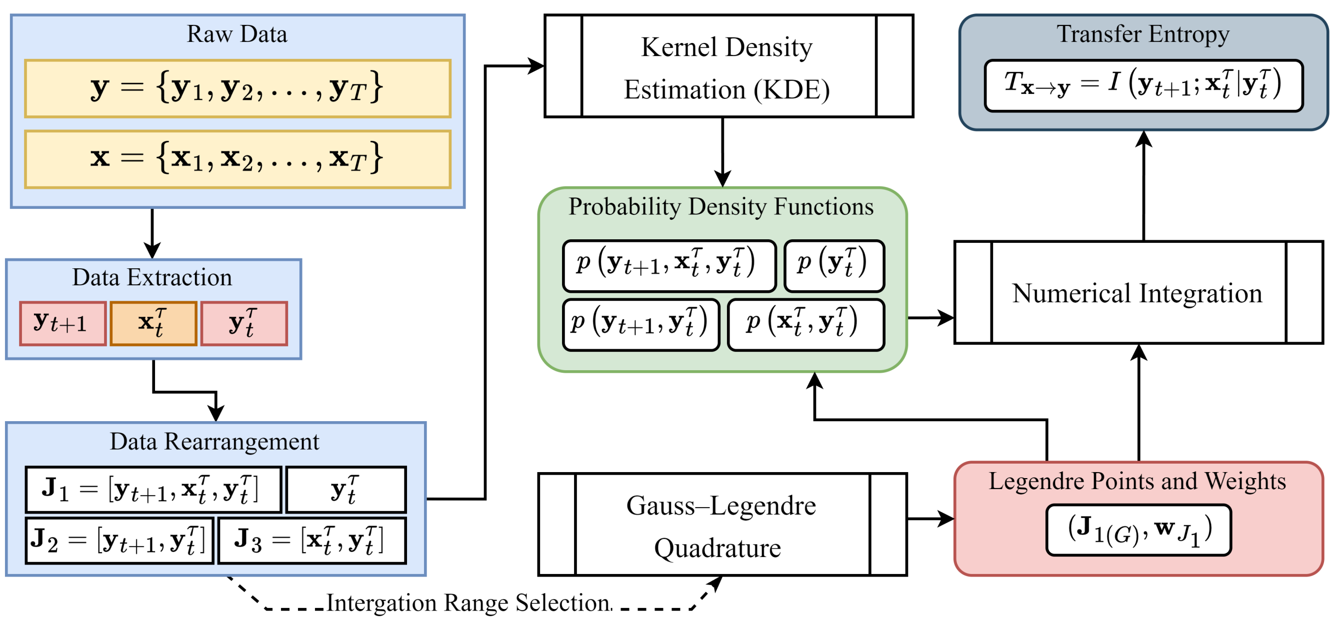


Figure 7. The Pipe Line of Transfer Entropy Estimation

4.2 Constrained Case with Single Observations per Time Step

In many real-world scenarios, the available raw data consist of only **one observation per unit of time**. For instance, the weekly stock price of a specific company in 1992 would typically be represented as a 1D sequence; obtaining multiple observations for the same system at the same instant is not feasible. The calculation process for such cases slightly diverges from that presented in subsection 4.1. Consider two sequential arrays $\mathbf{x}$ and $\mathbf{y}$, each of length T:

$$\mathbf{x} = \{x_1, x_2, \ldots, x_T\},$$
$$\mathbf{y} = \{y_1, y_2, \ldots, y_T\}.$$

Here, x_t and y_t are scalars, rendering $\mathbf{x}$ and $\mathbf{y}$ as 1D arrays of size T. In this scenario, the terms t, τ, and h denote the start time, the shift lag, and the window length, respectively. Two subsequences are extracted from the original sequences; for $\mathbf{y}$, these subsequences, $\mathbf{y}_p$ and $\mathbf{y}_f$, are defined as follows:

$$\mathbf{y}_p = \{y_t, y_{t+1}, \ldots, y_{t+h}\},$$
$$\mathbf{y}_f = \{y_{t+\tau}, y_{t+\tau+1}, \ldots, y_{t+\tau+h}\}.$$

A similar extraction process is applied to the sequence $\mathbf{x}$ to obtain $\mathbf{x}_p$. The transfer entropy is quantified as the reduction in entropy of $\mathbf{y}_f$ when accounting for both $\mathbf{y}_p$ and $\mathbf{x}_p$:

$$\begin{aligned} T_{\mathbf{x}\to\mathbf{y}}(t,\tau,h) &= H(\mathbf{y}_f|\mathbf{y}_p) - H(\mathbf{y}_f|\mathbf{y}_p,\mathbf{x}_p) \\ &= H(\mathbf{y}_f,\mathbf{y}_p) - H(\mathbf{y}_p) - H(\mathbf{y}_f,\mathbf{y}_p,\mathbf{x}_p) + H(\mathbf{y}_p,\mathbf{x}_p) \end{aligned} \quad (5)$$

And, in the sense of mutual information:

$$\begin{aligned} & T_{\mathbf{x}\to\mathbf{y}}(t,\tau,h) = I(\mathbf{y}_f;\mathbf{x}_p|\mathbf{y}_p) \\ & = \int_{\mathbf{y}_f}\int_{\mathbf{x}_p}\int_{\mathbf{y}_p} p(\mathbf{y}_f,\mathbf{x}_p,\mathbf{y}_p)\log\left(\frac{p(\mathbf{y}_p)p(\mathbf{y}_f,\mathbf{x}_p,\mathbf{y}_p)}{p(\mathbf{y}_f,\mathbf{y}_p)p(\mathbf{x}_p,\mathbf{y}_p)}\right) d\mathbf{y}_f d\mathbf{x}_p d\mathbf{y}_p \\ & = \int_{\mathbf{y}_f}\int_{\mathbf{x}_p}\int_{\mathbf{y}_p} p(\mathbf{y}_f,\mathbf{x}_p,\mathbf{y}_p)\left\{\log(p(\mathbf{y}_p)) + \log(p(\mathbf{y}_f,\mathbf{x}_p,\mathbf{y}_p)) - \log(p(\mathbf{y}_f,\mathbf{y}_p)) - \log(p(\mathbf{x}_p,\mathbf{y}_p))\right\} d\mathbf{y}_f d\mathbf{x}_p d\mathbf{y}_p \end{aligned} \quad (6)$$

Equation 5 and Equation 6 offer an alternative approach, particularly suitable for cases with limited data availability, to quantify information transfer. Figure 8 shows the concept of transfer entropy in the constrained case.

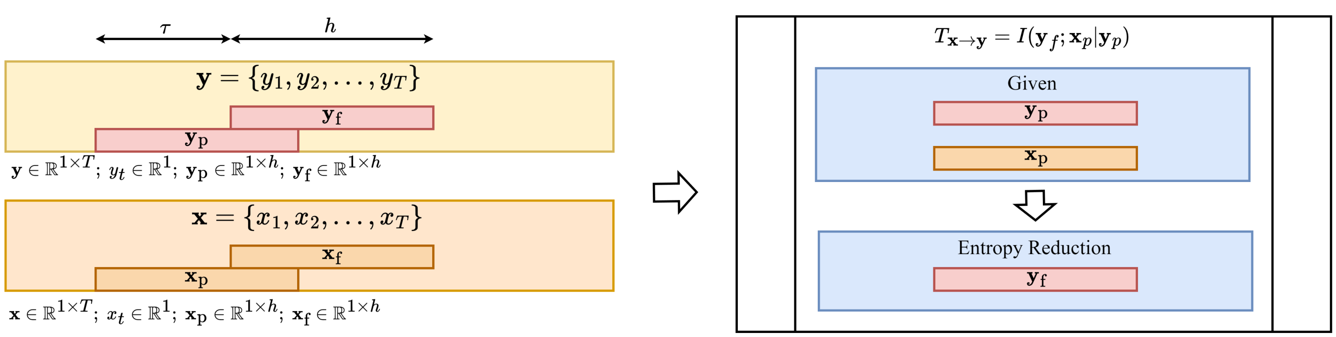


Figure 8. Concept of Transfer Entropy in Constrained Case

It's important to carefully consider the selection of τ and h in the constrained scenario depicted in Figure 8. Typically, $\mathbf{y}_f$ and $\mathbf{y}_p$ (as well as $\mathbf{x}_f$ and $\mathbf{x}_p$) should either partially overlap (if $h > \tau$) or be non-overlapping (if $h = \tau$). Avoiding $h < \tau$ is advisable as it creates a gap between past and future states, which can lead to inaccurate interpretations of the transfer entropy by disrupting the sequential continuity of the data.

Guideline for Choosing Between Scenarios

1. For scenarios where the number of observations is substantially larger than the sequence length ($N \gg T$), the general case is more suitable as there are sufficient observations to accurately compute transfer entropy.

2. In cases of very long sequence lengths (e.g., $T \to \infty$), or when there are relatively few observations per time step (e.g., $N = 1$ or $N \ll T$), the constrained case is preferable.

4.3 Numerical Experiments of Transfer Entropy

In the present section, random variable sequences are generated using different approaches, and the transfer entropy between them is measured. The ideal scenario described in subsection 4.1 is adopted, where it is possible to obtain multiple observations at each single step. Hence, a single feature sequence is a two-dimensional array with the shape $\mathbb{R}^{N\times T}$, where N and T denote the sample number and the sequence length (number of steps), respectively.

4.3.1 Sequences Formulation

A sequence $\mathbf{x}$ is generated, with 100 observations over 20 steps ($\mathbb{R}^{100\times 20}$), using the sigmoid function across the range $[-1, 10]$ plus random noise that increases with the step index. From the sequence $\mathbf{x}$, four sequences and one independent sequence are generated. They are as follows:

1. $\mathbf{y_1}$: Element-wise negative correlation, each $\mathbf{y_1}_{(t)}$ is directly related to $\mathbf{x}_{(t)}$, without historical correlation.

$$\mathbf{y_1}_{(t)} = -\mathbf{x}_{(t)} - 0.625\varepsilon t$$

2. $\mathbf{y_2}$: Linear historical dependence, where $\mathbf{y_2}_{(t)}$ depends on $\mathbf{x}_{(t-1)}$ and $\mathbf{y_2}_{(t-1)}$ at the previous step.

$$\mathbf{y_2}_{(t)} = 2\mathbf{x}_{(t-1)} - 0.002\mathbf{x}_{(t-1)}\mathbf{y_2}_{(t-1)} + 5|\varepsilon|$$

3. $\mathbf{y_3}$: Quadratic historical dependence, with $\mathbf{y_3}_{(t)}$ being a function of $\mathbf{x}^2_{(t-1)}$, $\mathbf{x}_{(t-1)}$, and $\mathbf{y_3}_{(t-1)}$ from the previous step.

$$\mathbf{y_3}_{(t)} = (\mathbf{x}^2_{(t-1)} - 0.2\mathbf{x}_{(t-1)}\mathbf{y_3}_{(t-1)} + 10|\varepsilon|)/t^2$$

4. $\mathbf{y_4}$: Cubic historical dependence, where $\mathbf{y_4}_{(t)}$ is a function of $\mathbf{x}^3_{(t-1)}$, $\mathbf{x}^2_{(t-1)}$, $\mathbf{x}_{(t-1)}$, and $\mathbf{y_4}_{(t-1)}$ from the previous step.

$$\mathbf{y_4}_{(t)} = (\mathbf{x}^3_{(t-1)} + \mathbf{x}^2_{(t-1)} - \mathbf{x}_{(t-1)}\mathbf{y_4}_{(t-1)} + 15|\varepsilon|)/t^3$$

5. $\mathbf{z}$: Independent sequence, where $\mathbf{z}_{(t)}$ is generated as a function of two distinct noise terms, with $\mathbf{z}_{(t)} = 5\bar{\varepsilon}^2 - 6\varepsilon$.

Where ε is Gaussian random noise with $\varepsilon \sim \mathcal{N}(0,1)$, and $\bar{\varepsilon}$ is uniform random noise with $\bar{\varepsilon} \sim \mathcal{U}(0,1)$. All sequences and their distribution at each step are detailed in Figure 9. And the graphical representation showing dependence between sequences is demonstrated in Figure 10.

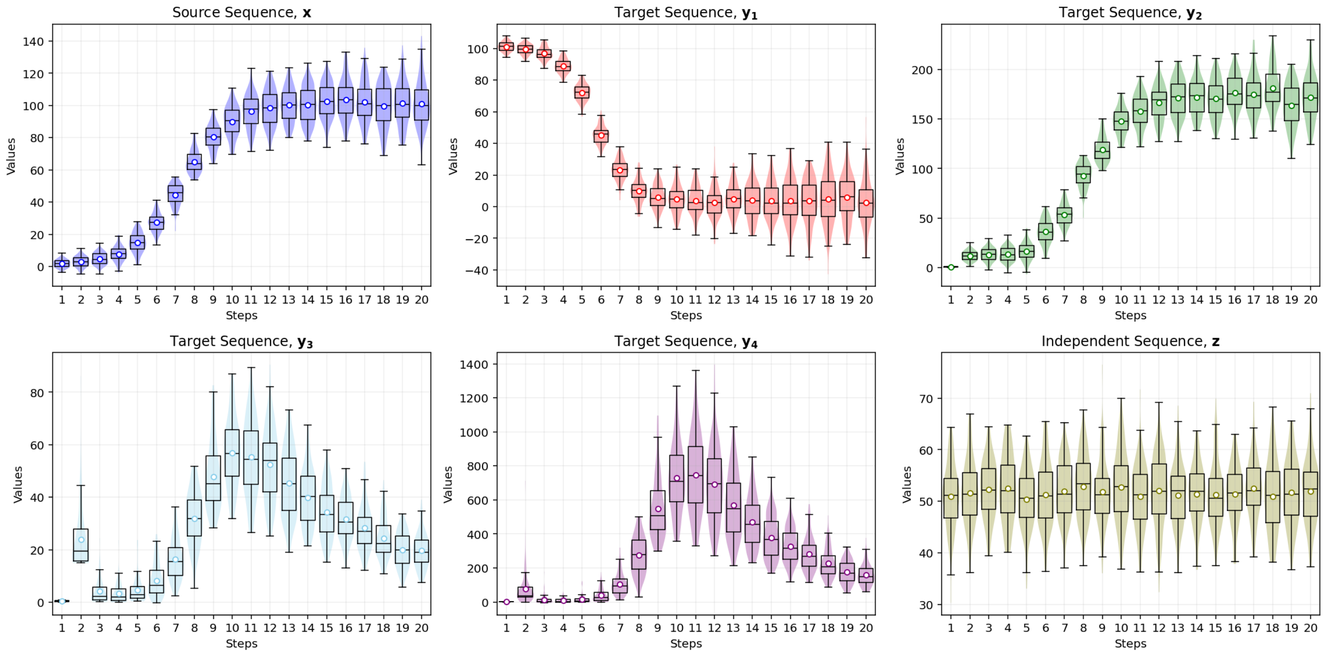


Figure 9. Sequences for Transfer Entropy Experiments

4.3.2 Experiment 1: Directionality, History Length, and Temporal Correlation Effects

To aviod confusion, the source and target are denoted as $\mathbf{a}$ and $\mathbf{b}$, respectively, which is different from the notations of previous section, i.e., $\mathbf{x}$ represents source and $\mathbf{y}$ represents target. Following the process outlined in Figure 7, the sequence $\mathbf{x}$ is always referred to as the source sequence, i.e., $\mathbf{a} = \mathbf{x}$. The other sequences are designated as target sequences, i.e., $\mathbf{b} \in \{\mathbf{y_1}, \mathbf{y_2}, \mathbf{y_3}, \mathbf{y_4}, \mathbf{z}\}$. Given that the sequence length $T = 20$, the number of observations for different history lengths τ varies. For example, for $\tau = 1$, there are $T - \tau = 19$ transfer entropy values estimated: $\{T_{\mathbf{x}\to\mathbf{y}}(t = 1, \tau = 1), \ldots, T_{\mathbf{x}\to\mathbf{y}}(t = 19, \tau = 1)\}$, where t denotes the observation step at which the information transfer to the next step is estimated. The average transfer entropy values across all observation steps are presented in Table 4 for $\tau = 1$ and $\tau = 2$. Additionally, the computational costs for estimating all transfer entropy values in both directions (from source to target and vice versa) for all sequence pairs are reported.

		$\tau = 1$		$\tau = 2$	
	$(\mathbf{a}\ ,\ \mathbf{b})$	$T_{\mathbf{a}\to\mathbf{b}}$	$T_{\mathbf{b}\to\mathbf{a}}$	$T_{\mathbf{a}\to\mathbf{b}}$	$T_{\mathbf{b}\to\mathbf{a}}$
Element-Wise	$(\mathbf{x}, \mathbf{y_1})$	0.0772	0.0788	0.2012	0.2013
Linear	$(\mathbf{x}, \mathbf{y_2})$	395.82	0.0751	5301.8	3.1181
Quadratic	$(\mathbf{x}, \mathbf{y_3})$	3.1849	0.0728	3.3025	0.1711
Cubic	$(\mathbf{x}, \mathbf{y_4})$	1.7302	0.0721	55.738	3.2279
Independent	$(\mathbf{x}\ ,\ \mathbf{z})$	0.0734	0.0772	0.1863	0.1883
Computational Cost (s)		8.6		2212.2	

Table 4. Average Transfer Entropy over all observation steps for Different correlation for varying τ, unit of transfer entropy is given as bits.

1. Transfer entropy(TE) is an asymmetric measure; that is, $T_{\mathbf{a}\to\mathbf{b}} \neq T_{\mathbf{b}\to\mathbf{a}}$. A large $T_{\mathbf{a}\to\mathbf{b}}$ does not necessarily imply a large $T_{\mathbf{b}\to\mathbf{a}}$. For linear, quadratic, and cubic historical dependence, as shown in Table 4, the average transfer entropy from the source $\mathbf{x}$ to the targets $\mathbf{y_1}$, $\mathbf{y_2}$, or $\mathbf{y_3}$ is consistently significantly larger than from the targets to the source. This directional dependence property provides insights into causality.

2. Transfer entropy may not effectively capture element-wise correlation, as demonstrated in Table 4. Sequence $\mathbf{z}$ is formulated independently of $\mathbf{x}$, whereas $\mathbf{y_1}$ exhibits an element-wise correlation with $\mathbf{x}$. Despite this, the average transfer entropy for the sequence pair $(\mathbf{a}, \mathbf{b}) = (\mathbf{x}, \mathbf{y_1})$ is only slightly higher than that for $(\mathbf{a}, \mathbf{b}) = (\mathbf{x}, \mathbf{z})$. Specifically, in the

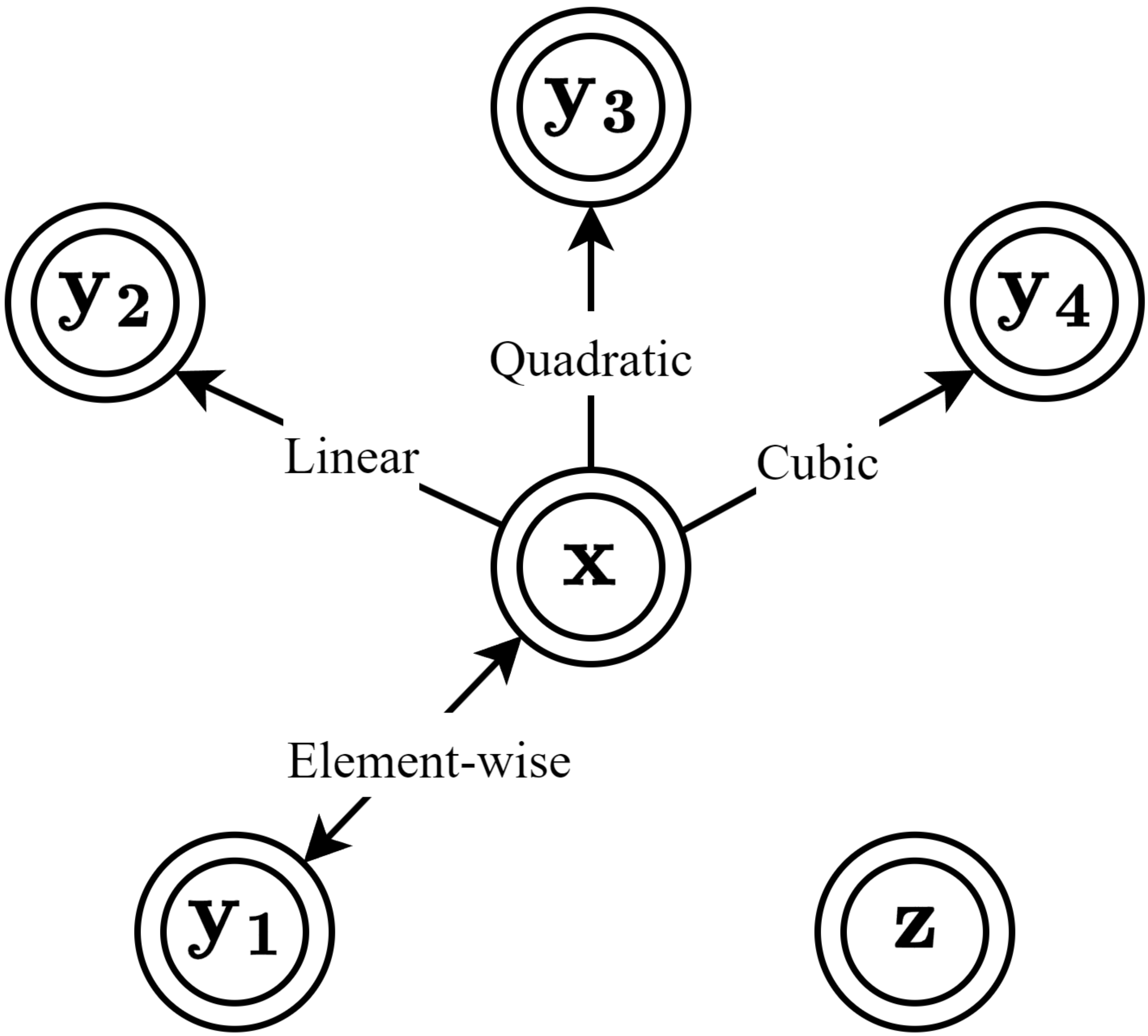
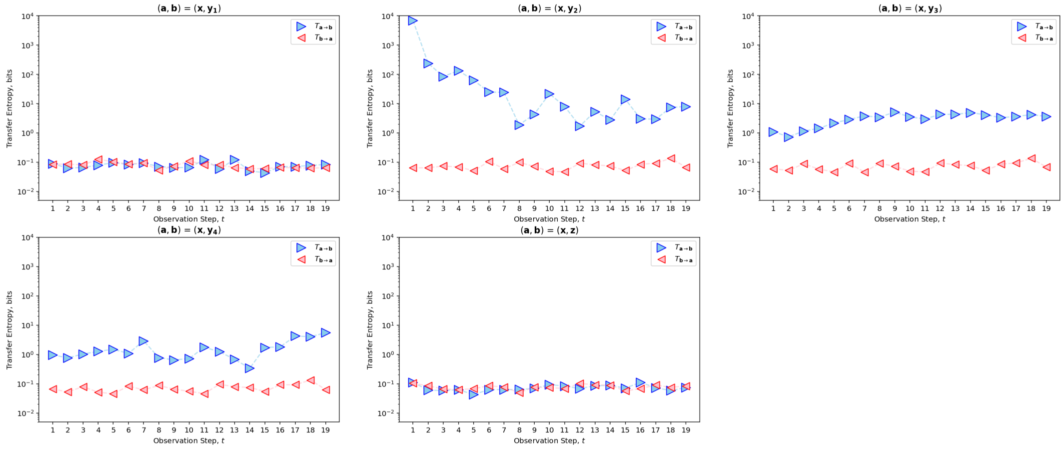
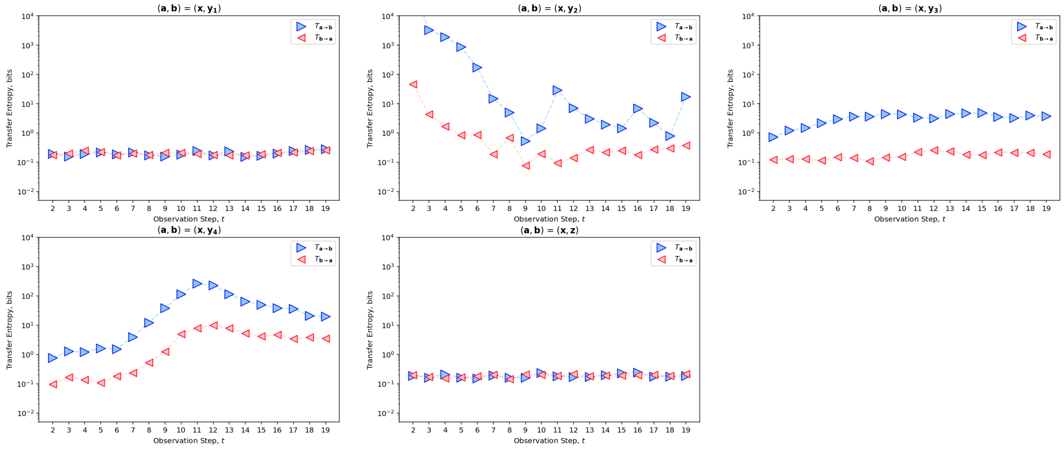


Figure 10. Graphical Demonstration of Sequences Dependence



(a) Transfer Entropy across all observation steps for $\tau = 1$.



(b) Transfer Entropy across all observation steps for $\tau = 2$.

Figure 11. Transfer Entropy across all observation steps with varying history length τ.

element-wise case, $T_{\mathbf{a}\to\mathbf{b}}$ is 5.17% higher than the independent case for $\tau = 1$, and 7.99% higher for $\tau = 2$. Despite these differences, the values are still on the same numerical level.

3. When the target sequence is formulated as a function of the past values of the source sequence, the transfer entropy values tend to be very high. This is evident in Table 4, where for cases with historical correlation, $T_{\mathbf{a}\to\mathbf{b}}$ is significantly larger than $T_{\mathbf{b}\to\mathbf{a}}$.

4. Increasing the τ value generally results in higher transfer entropy values. This is expected when there is a historical dependence between sequences, as including more historical data tends to reduce the uncertainty of the target sequence, thereby increasing transfer entropy. Interestingly, this increase is also observed in element-wise and independent sequence cases, likely due to statistical fluctuations or noise, which can yield non-zero values even in the absence of a true underlying reltaion.

5. The computational cost of transfer entropy estimation increases exponentially with the history length, τ, due to the nature of joint probability density functions. The key array, $\mathbf{J_1}$, comprises three components: $\mathbf{b}_{t+1}$, $\mathbf{a}_t^\tau$, and $\mathbf{b}_t^\tau$ (corresponding to $\mathbf{y}_{t+1}$, $\mathbf{x}_t^\tau$, and $\mathbf{y}_t^\tau$ in Figure 7). As τ increases, both $\mathbf{a}_t^\tau$ and $\mathbf{b}_t^\tau$ expand to include more historical data, thus increasing the dimensionality of $\mathbf{J_1}$. For instance, with 16 Gauss points per dimension and $\tau = 1$, the estimation requires 16^3 Gauss points. With $\tau = 2$, this number rises to 16^5. If $\tau = 3$, it further escalates to 16^7. Generally, $n^{(1+2\tau)}$ points are needed for the joint density estimation, where n is the number of Gauss points per dimension. This exponential growth in required integration points contributes to the significant computational cost for higher values of τ. This potentially explains why most examples in the literature use the constrained scenario (subsection 4.2) to illustrate transfer entropy.

4.3.3 Experiment 2: Implicit Historical Information Flow Detection

Recall that sequence $\mathbf{y_1}$ is derived from $\mathbf{x}$ through element-wise operations, resulting in low transfer entropy between them in both directions. Sequences $\mathbf{y_2}$, $\mathbf{y_3}$, and $\mathbf{y_4}$, however, are derived from $\mathbf{x}$ via operations that introduce historical dependence. It is intriguing to investigate whether $\mathbf{y_1}$ can inform future predictions for $\mathbf{y_2}$, $\mathbf{y_3}$, and $\mathbf{y_4}$, as well as whether knowledge of $\mathbf{y_2}$ can reduce the historical uncertainty of $\mathbf{y_3}$ and $\mathbf{y_4}$, and so forth. In this section, the history length is set to $\tau = 1$. Following the approach outlined in Table 4, the average transfer entropy over all observation steps for the six unique pairs within the sequence set $\{\mathbf{y_1}, \mathbf{y_2}, \mathbf{y_3}, \mathbf{y_4}\}$ is presented in Table 5.

$(\mathbf{a}\ ,\ \mathbf{b})$	$T_{\mathbf{a}\to\mathbf{b}}$	$T_{\mathbf{b}\to\mathbf{a}}$
$(\mathbf{y_1}, \mathbf{y_2})$	0.0743	0.0770
$(\mathbf{y_1}, \mathbf{y_3})$	0.0628	0.0724
$(\mathbf{y_1}, \mathbf{y_4})$	0.0627	0.0725
$(\mathbf{y_2}, \mathbf{y_3})$	0.0444	0.0435
$(\mathbf{y_2}, \mathbf{y_4})$	0.0609	0.0644
$(\mathbf{y_3}, \mathbf{y_4})$	0.0746	0.0733

Table 5. Average Transfer Entropy for Implicit Historical Dependence

Table 5 reveals that the implicit historical dependence captured by the average TE for each pair is notably weak. Most of the average values presented in Table 5 are on the same level as those for the independent pair case shown in Table 4. Intriguingly, some values are even lower, suggesting an average information flow that is insufficiently strong to indicate significant historical dependence for implicit/indirect historical dependent pairs.

4.3.4 Permutable Test of Transfer Entropy

Given the biases arising from finite data lengths and the nature of numerical estimation approaches (such as numerical integration for continuous variables discussed in this document, or discretizing data into histograms to calculate conditional mutual information), it is uncommon to observe zero TE values, even in the absence of historical dependence. A commonly adopted approach is to establish a threshold; if the transfer entropy falls below this threshold, it is directly set to zero. The permutation test is utilized to determine this threshold. Consider transfer entropy from $\mathbf{a}$ to $\mathbf{b}$, with each array having the shape $(N \times T)$. The steps are as follows:

1. Generate a set of R permuted sequential arrays, denoted by $\mathbf{a'}_{(r)}$, each formulated by randomly shuffling $\mathbf{a}$ in the temporal dimension.

2. For a specific step t and a given history length τ, calculate the transfer entropy $T_{\mathbf{a}\to\mathbf{b}}$, as well as the transfer entropy for each permuted sequence. This yields the permuted transfer entropy set $\mathbf{TE_p} = \{T_{\mathbf{a'}_{(1)}\to\mathbf{b}}, T_{\mathbf{a'}_{(2)}\to\mathbf{b}}, \ldots, T_{\mathbf{a'}_{(R)}\to\mathbf{b}}\}$.

3. Estimate the cumulative distribution function (CDF) of $\mathbf{TE_p}$, denoted as Φ.

4. Select a cumulative probability threshold ρ. Then, $T_{\mathbf{a}\to\mathbf{b}}$ is set to zero if $\Phi(T_{\mathbf{a}\to\mathbf{b}}) \leq \rho$; otherwise, it retains its original estimation.

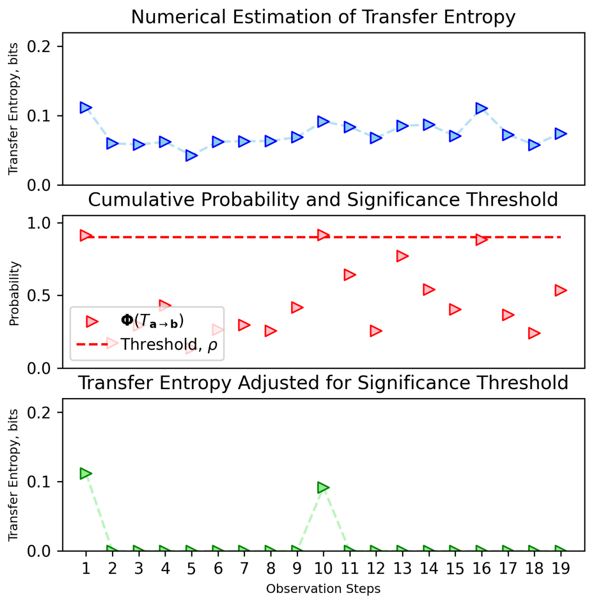
Four sequences $\{\mathbf{x}, \mathbf{y_1}, \mathbf{y_2}, \mathbf{y_3}\}$ are used, while instead of considering all pairs in both directions, only 4 cases are considered. The number of permuted sequences and the probability threshold are set up as $R = 50$ and $\rho = 0.9$, respectively. Results are given in Figure 12, and the illustrations are listed according to sequence relation in Figure 10:

1. Figure 12a: Transfer entropy $T_{\mathbf{x}\to\mathbf{z}}$ for the independent pair $(\mathbf{x}, \mathbf{z})$. Sequence $\mathbf{z}$ is generated independently of all other sequences; hence, statistical insignificance for transfer entropy is expected.

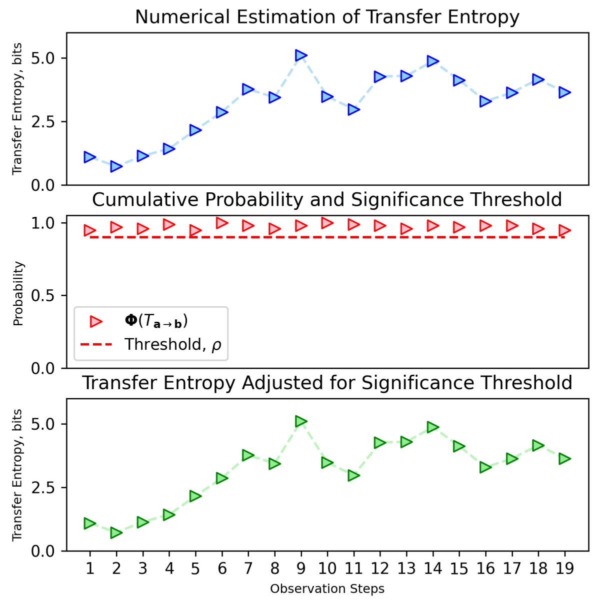
2. Figure 12b: Transfer entropy $T_{\mathbf{x}\to\mathbf{y_3}}$ for the quadratic dependent pair $(\mathbf{x}, \mathbf{y_3})$. In the sense of a directed graph in Figure 10, the edge $E(\mathbf{x}, \mathbf{y_3}) = \mathbf{x} \to \mathbf{y_3}$ exists; therefore, transfer entropy in this case, throughout all time steps, is always expected to be statistically significant.

3. Figure 12c: Transfer entropy $T_{\mathbf{y_1}\to\mathbf{y_2}}$ for the indirect pair $(\mathbf{y_1}, \mathbf{y_2})$. There is no direct relation between $\mathbf{y_1}$ and $\mathbf{y_2}$; in terms of graph sense, $\mathbf{y_2}$ is 'connected' with $\mathbf{y_1}$ through $\mathbf{x}$, as there is an element-wise dependence between $\mathbf{y_1}$ and $\mathbf{x}$.

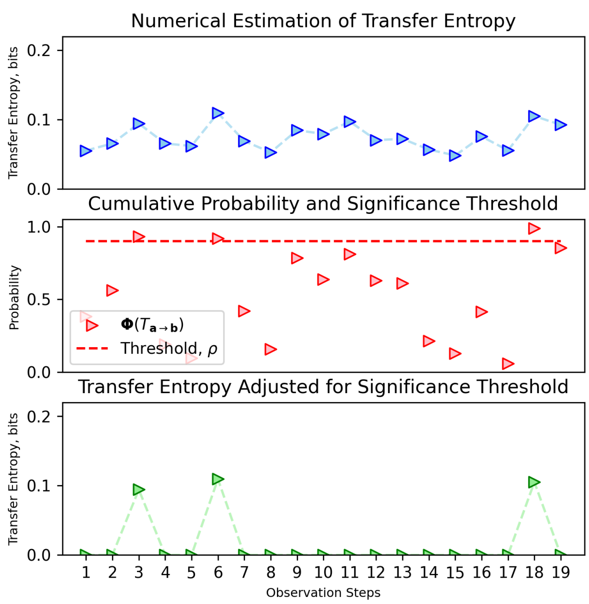
4. Figure 12d: Transfer entropy $T_{\mathbf{y_1}\to\mathbf{y_3}}$ for the indirect pair $(\mathbf{y_1}, \mathbf{y_3})$. Similar to the pair $(\mathbf{y_1}, \mathbf{y_2})$, the edge $E(\mathbf{y_1}, \mathbf{y_3})$ does not exist, while they are linked through $\mathbf{x}$.



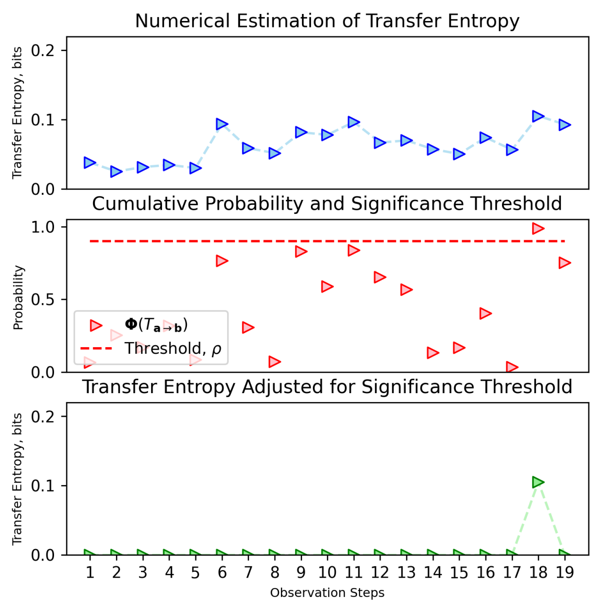
(a) Independence, $T_{\mathbf{x}\to\mathbf{z}}$



(b) Quadratic Dependence, $T_{\mathbf{x}\to\mathbf{y_3}}$



(c) Indirect Dependence, $T_{\mathbf{y_1}\to\mathbf{y_2}}$



(d) Indirect Dependence, $T_{\mathbf{y_1}\to\mathbf{y_3}}$

Figure 12. Adjusted Transfer Entropy based on Permutable Test

The results from Figure 12 indicate that transfer entropy may not effectively identify indirect historical dependencies. In such cases, raw transfer entropy values greater than zero are likely attributable to statistical noise or the accumulation of estimation errors. Therefore, a statistical test is essential to 'purify' these values by distinguishing significant signals from noise. However, it's important to note that this purification process can be computationally expensive, as it necessitates repeated calculations of transfer entropy for permutation testing.

Alternatively, a hypothesis test can be utilized to assess the statistical significance of the transfer entropy value, $T_{\mathbf{a}\to\mathbf{b}}$, relative to the distribution of values obtained from the permuted dataset. The null hypothesis is established as:

$$H_0 : T_{\mathbf{a}\to\mathbf{b}} = 0,$$
$$H_1 : T_{\mathbf{a}\to\mathbf{b}} \neq 0.$$

A statistical test, such as the t-test, is then employed to compare the actual observed transfer entropy against the distribution from the permuted sets to determine whether the null hypothesis can be rejected.

5 Conclusions and Insights

5.1 Conclusions

Based on the results from subsubsection 4.3.2, subsubsection 4.3.3 and subsubsection 4.3.4, several conclusions about transfer entropy(TE) can be drawn:

1. Transfer entropy (TE) is asymmetric, providing directional insights into information flow between sequences.
2. TE values tend to be significantly higher when there is direct historical or temporal dependence between sequences. However, TE may not effectively capture indirect historical dependence.
3. Low TE values do not necessarily indicate the absence of information flow. Additional tests, such as the permutable test or statistical tests, are often required for further verification.
4. In the most general case, the value of Transfer Entropy (TE) depends on the observed step t, the past history length τ, and the future window length h. Without further simplification, including the permutation test in the entire process makes using $\tau > 1$ and $h > 1$ impractical, as the required computational time grows exponentially, leading to prohibitive computational costs. Consequently, algorithms designed for constrained scenarios, as discussed in subsection 4.2, are more commonly used in practical applications due to their feasibility and lower computational requirements.

References

1. Moon, Y.-I., Rajagopalan, B. & Lall, U. Estimation of mutual information using kernel density estimators. *Phys. Rev. E* **52**, 2318 (1995).

2. Belghazi, M. I. *et al.* Mutual information neural estimation. In *International conference on machine learning*, 531–540 (PMLR, 2018).

3. Chan, C. *et al.* Neural entropic estimation: A faster path to mutual information estimation. *arXiv preprint arXiv:1905.12957* (2019).

4. Edition, F., Papoulis, A. & Pillai, S. U. *Probability, random variables, and stochastic processes* (McGraw-Hill Europe: New York, NY, USA, 2002).

5. Barnett, L., Barrett, A. B. & Seth, A. K. Granger causality and transfer entropy are equivalent for gaussian variables. *Phys. review letters* **103**, 238701 (2009).

6. Majda, A. J. & Chen, N. Model error, information barriers, state estimation and prediction in complex multiscale systems. *Entropy* **20**, 644 (2018).

7. Kim, P., Rogers, J., Sun, J. & Bollt, E. Causation entropy identifies sparsity structure for parameter estimation of dynamic systems. *J. Comput. Nonlinear Dyn.* **12**, 011008 (2017).

8. Shahsavari Baboukani, P., Graversen, C., Alickovic, E. & Østergaard, J. Estimating conditional transfer entropy in time series using mutual information and nonlinear prediction. *Entropy* **22**, 1124 (2020).

6 Appendix: Relation between Entropy-Based Quantities

Given the operations and hierarchy between entropy-based quantities:

Operations:

1. $H(X,Y) = H(X) + H(Y|X) = H(Y) + H(X|Y)$ (Definition of conditional entropy)
2. $I(X;Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$ (Definition of mutual information)
3. $I(X;Y) = H(X) + H(Y) - H(X,Y)$ (Alternative expression for mutual information)

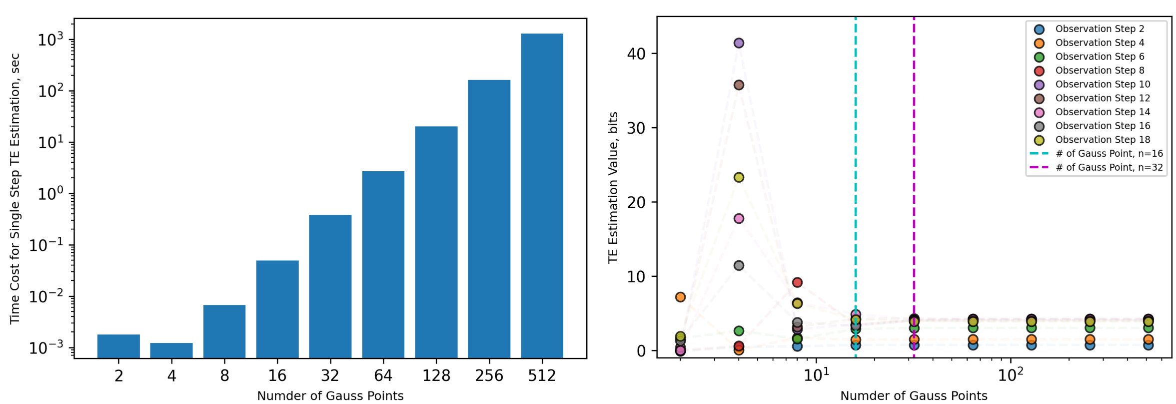
Hierarchy Relationships:

1. $H(X) + H(Y) \geq H(X,Y)$ (Subadditivity of entropy)
2. $H(X,Y) \geq \max\{H(X), H(Y)\}$ (Joint entropy is at least as large as any of the marginals)
3. $H(X|Y) \leq H(X)$ (Knowledge of Y cannot increase uncertainty about X)
4. $I(X;Y) \leq H(X)$ (Information shared between X and Y cannot exceed the entropy of X)
5. $I(X;Y) \geq 0$ (Mutual Information is strictly non-negative)

1. The **operations** are universally true for both continuous and discrete variables, providing a consistent framework for calculating and relating informational measures. However, the **hierarchy relationships**, except for the non-negativity of Mutual Information, are likely not hold for continuous variables.
2. When estimating these information measures for continuous variables, it's recommended to use direct numerical methods on the original definitions. This approach helps avoid potential errors that can happen when using step-by-step methods, which might enlarge small inaccuracies.

7 Appendix: Number of Gauss Points

In order to demonstrate how to set the appropriate number of Gauss-Legendre points, n, sequences $\mathbf{x}$ and $\mathbf{y}_3$ are selected to conduct a convergence test for the transfer entropy (TE) value with respect to the number of Gauss-Legendre points and weights. The steps $t \in \{2,4,6,8,10,12,14,16,18\}$ and Gauss point numbers $n \in \{2,4,8,16,32,64,128,256,512\}$ are adopted for the convergence test. The average computational costs over the testing steps for each number of Gauss points are presented in Figure 13a, and the transfer entropy values for each setting at each number of Gauss points are shown in Figure 13b. It is shown that n should be at least greater than 16 to avoid compromising the accuracy of numerical integration. In this document, $n = 16$ is chosen to consider a trade-off balance between TE accuracy and computational cost.



(a) Computational Cost with Increased Gauss Points

(b) TE Convergence with Increased Gauss Points

Figure 13. Test of TE Cost and Convergence with Number of Gauss Points

8 Appendix: Personal Murmur

To summarize transfer entropy(TE), this section lists the advantages:

1. A very cool advanced technique, especially when dealing with continuous sequential variables.
2. Provides step-wise details, which is its most significant ingredient.
3. Digestion of affiliated benefits: Learning transfer entropy for continuous variables imparts numerous techniques, such as kernel density estimation, numerical integration, and entropy estimation. Especially from an implementation perspective, these experiences are invaluable.

And the disadvantages:

1. Very steep learning curve, similar to mutual information. Furthermore, in the usual literature, there is a huge gap between the general case from the original mathematical definition (Figure 6) and the constrained case with simplification (Figure 8), without any proper explanation, which can be really confusing for a beginner.
2. Computational cost for TE estimation is usually very high due to kernel density estimation and numerical integration. Without further pre-assumption or simplification, e.g., estimate entropy based on covariance approximation (subsection 3.3), it's not feasible to expand the degree of freedom (e.g., incorporate more past historical data by increasing τ) on running estimation.